# Paul Shenoda

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## Professional Summary


AI-first software engineer who ships production AI systems end to end: RAG pipelines, multi-agent workflows and full-stack apps on Azure. Uses Claude Code and Codex throughout the development lifecycle while owning architecture, testing and verification. Background includes 8 months on a safety-critical industrial platform at Skyjack and a Computer Engineering and Management degree from McMaster University.


## Technical Skills

- **AI & LLM**: RAG, embeddings, vector search (Azure AI Search), prompt engineering, multi-agent workflows (LangGraph), LLM evaluation (RAGAS), LangSmith tracing, Claude Code, Codex
- **Languages & Frameworks**: Python, C/C++, JavaScript, TypeScript, HTML/CSS, SQL, FastAPI, LangChain, LangGraph, React, React Native, Next.js
- **Cloud & DevOps**: Azure (OpenAI, AI Search, Container Apps/Jobs), Docker, Git, GitHub Actions
- **Platforms & Tools**: Node.js, PostgreSQL, Prisma, SQLite, MySQL, Raspberry Pi, Google Test, pytest

## Certifications

- **Anthropic** – Claude Certified Developer, Foundations (CCDV-F), July 2026
- **Anthropic** – Claude Certified Architect, Foundations (CCAR-F), In Progress
- **Microsoft** – Azure AI Fundamentals (AI-900), January 2026

## Work Experience

**Embedded Software Engineer Co-op** *Jan 2023 – Aug 2023*
*Skyjack – Guelph, ON*

- Extended from an initial 4-month term to a full 8 months based on performance; later received a verbal offer to return after graduation and chose to pursue software and AI-focused roles instead
- Developed and validated 30+ **CANBus** message protocols across isolated control boxes, confirming correct LED and alarm trigger behavior for inter-system communication
- Reduced on-machine field testing time by ~40% by provisioning 4 in-office hardware control boxes and updating system schematics, enabling **CANBus** fault validation directly at developers' desks
- Maintained sprint velocity by providing daily cross-functional support to QA, software, and electrical teams during **Agile** sprints, unblocking core developers on technical tasks

## Technical Project Experience

**RAG Document Chatbot** | *Python, LangChain, FastAPI, Azure OpenAI* *Mar 2026 – Apr 2026*

- Started as a capstone with an industry partner: interviewed the client to scope the problem, then built the full **RAG** pipeline independently after the course ended
- Used **LangChain** and **Azure OpenAI** for grounded document Q&A, with **Azure AI Search** as the vector store for sub-second retrieval; deployed via **Docker** and **GitHub Actions** to **Azure Container Apps**
- Evaluated the chatbot with a 30-question RAGAS suite spanning 4 vendors, scoring 91% faithfulness (**GPT-4o** judge, average across 3 runs), 87% context recall and 85% answer relevancy

**AI Crypto Trading Bot** | *Python, LangGraph, Azure Container App Jobs* *Apr 2026 – June 2026*

- Built a **multi-agent trading system**, coordinating Sentiment, Risk Manager, and Trader agents through **LangGraph**'s supervisor pattern, defining the agent roles and building the architecture with Claude
- Fixed a hallucination bug where **GPT-4o-mini** misread close decimals as correct (69.5 read as greater than 70) by tracing its reasoning in **LangSmith** and moving comparisons to deterministic Python

- Resolved a production freeze that halted trading for 28 cycles during a live 4.3% rally by adding defensive type casting and alerting, after migrating the bot from testnet to live **Kraken** trading with real capital

**Background Job Runner** | *Python, FastAPI, PostgreSQL, React, Claude Code* | GitHub *July 2026*

- Built an end-to-end job orchestration engine with collision-free handle recycling, capped concurrency, and crash-durable recovery, passing the assessment's hardest-weighted correctness tests, by combining atomic Postgres allocation with lease-based claiming and heartbeat renewal
- Scored **9.6/10** on a 28-criteria take-home technical assessment by building a documented **REST**/**SSE** API with per-user auth and a **pytest** suite covering a real process-kill restart test

**Capstone: Watch Timegrapher** | *Python, C++, Teensy MCU* *Sept 2025 – Apr 2026*

- Owned the DSP & Algorithm module end-to-end: prototyped in **Python**, then refactored into stateful **C++** on a **Teensy** MCU, hitting **≤1 ms beat error** and **≤10% amplitude accuracy** against spec
- Verified correctness with **unit tests** on filter state, rectifier behavior, and DFT accuracy; validated the full pipeline against live piezo sensor data

## Volunteer & Community Projects

**Bible Reading Tracker** | *Next.js, TypeScript, Prisma, Auth.js, Claude Code* *July 2026 – Present*

- Built a full-stack app using **Claude Code** and **Claude Design**, from database schema to login to admin tools, to help high school students track daily Bible reading; now used by **68 registered students**, with about 30 completing their reading every day
- Set up a **4-role permission system** with grade-scoped access, streaks and points to keep students motivated, and **4 scheduled jobs** for nightly streak resets, event lifecycle, and push and event reminders
- Self-hosted at no cost using **Docker** and a persistent tunnel, with data export and an activity log built for handoff to non-technical church leadership

**Tournament Bracket Manager V2** | *Google Sheets, JavaScript* *June 2026*

- Built the Google Sheets backend and live bracket logic for a lightweight rebuild of an earlier FastAPI/React Native app, with a friend building the front end; removed the need for any hosting
- Used by **460+ students** over a live July 2026 tournament weekend

## Education

**McMaster University** *June 2026*

*B.Eng. Computer Engineering and Management*